SSRNet: A Lightweight Successive Spatial Rectified Network With Noncentral Positional Sampling Strategy for Hyperspectral Images Classification

Dongmei Song, Changlong Yang, Bin Wang, Jie Zhang, Han Gao, *Member, IEEE*, and Yunhe Tang

***Abstract*— Deep learning methods have been proved outperforming the traditional methods in the field of hyperspectral image classification (HSIC). However, in pursuit of higher accuracy, HSIC networks have become deeper and more complex, resulting in excessive parameters and computational cost. To deploy neural networks on small platforms such as mobile or embedded devices, many studies have focused on lightweight HSIC networks. Currently, these works are dominated by patch-based networks, which suffer from the low accuracy caused by lightweight scale and slow inference speed derived from structural deficiencies of such networks. It is worth noting that full convolutional networks (FCNs) are able to achieve fast inference, but they tend to consume massive memory. To this end, this article proposes a novel lightweight HSIC method, which consists of a successive spatial rectified network (SSRNet) and a noncentral positional sampling (NCPS) strategy. SSRNet is composed of a local channel attention-based spectral FCN and several separable atrous spatial pyramid modules (SASPMs). These shallow subnetworks are concatenated together to progressively optimize their outputs by successive spatial rectified learning. For decreasing memory access cost (MAC), SSRNet makes little patches as input to perform patch-wise pixel-to-pixel learning. After training, SSRNet is able to adapt to any size of hyperspectral images and complete fast inference of the full image directly. In particular, the NCPS sampling strategy enables all labeled pixels to equally traverse all spatial positions of each training patch through the positional shift sampling, which effectively alleviates the sparse problem of hyperspectral semantic labels. Experiments upon three public benchmark datasets indicate that SSRNet is comparable to the state-of-the-art methods in classification accuracy with less than 0.15 M parameters and only occupies less than 10-MB memory for single forward computation. Moreover, SSRNet behaves significantly superior to traditional patch-based networks in terms of the inference speed. The source codes can be available from the website of https://github.com/Pancakerr/HSIC-platform.**

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***Index Terms*— Fully convolutional network (FCN), hyperspectral image classification (HSIC), lightweight, local channel attention (LCA), spatial rectified learning.**

I. Introduction

HYPERSPECTRAL remote sensing images (HSIs) contain the spectral information of ground objects in hundreds of bands, which can provide a complete spectral profile of each pixel and thus enable the fine-grained identification and detection. Therefore, HSIs are widely used in the fields of vegetation, soil, minerals, and agriculture [1], [2], [3], [4]. In the past decade, traditional machine learning methods, such as support vector machine (SVM) [5], K-nearest neighbor (KNN) [6], extended morphological profile (EMP) [7], and markov random field (MRF) [8], have been mainly used to fulfill hyperspectral image classification (HSIC) tasks. However, these methods mainly rely on handcraft features, lack flexibility, and have limited classification accuracy. With the explosive development of neural networks technology in recent years, many classical neural networks in computer vision, such as convolutional neural network (CNN) [9], recurrent neural network (RNN) [10], stacked autoencoder (SAE) [11], deep belief network (DBN) [12], and generative adversarial network (GAN) [13], have been gradually introduced into the field of HSIC. Many studies have shown that these data-driven deep learning methods dramatically outperform traditional methods in terms of accuracy and generalization performance. Among them, as an efficient hierarchical image feature representation learning framework, CNN has gained widespread popularity in the HSI community.

Nowadays, the design of CNN for HSIC exhibits the trend of multidimensionality, multi-information, multimodularity, and deepness. The multidimensionality is mainly reflected in the application of 1-D-CNN [14], 2-D-CNN [15], 3-D-CNN [16], and even mixed-dimension CNN [17], while the multi-information is reflected by the joint spectral–spatial features of HSI [18], [19]. In addition, many novel computer vision technologies have been continuously applied to HSIC, among which the attention module is widely used due to its ability to guide the network to focus on more important information. For instance, the convolutional block attention module (CBAM) is adopted for automatically driving the network to concern more significant bands and spatial ranges according to the data distribution [20], [21]. Roy et al. [22] introduced an adaptive spatial–spectral kernel attention module into HSIC, enabling the network to automatically choose the proportion

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of spatial and spectral information during feature fusion. In conclusion, the diversity of dimensions, the completeness of information, and the intricacy of modules have increasingly made network deeper and more complex. However, it is obviously difficult for small platforms such as mobile or embedded devices to deploy these complex hardware-dependent networks so that there is an increasingly urgent demand for accurate lightweight classification networks.

However, there are two important issues that need to be addressed to achieve a lightweight network. The first is to reduce the parameter scale while ensuring accuracy. Generally, a complicated CNN can be lightened via replacing the standard convolution by using simplified convolutional modules such as grouped convolution, depth-wise separable convolution [23], and Ghost convolution [24]. Although these designs can reduce some of the network parameters, they still maintain a certain depth to guarantee accuracy. In other words, the structure needs to be further optimized. The second is to improve the inference speed of the network while restricting memory access cost (MAC). Although lightweight networks outperform traditional networks in terms of speed, there is still a big gap from the desired real-time inference. This can be attributed to the patch-based structure, which must generate patches corresponding to all pixels, and then classify them one by one, which causes lots of redundant calculations and thus slows down the inference speed. As a typical representative of end-to-end network, fully convolutional network (FCN) [25] only contains convolutional operations without any fully connected layers and can accept images of arbitrary size as inputs, thereby realizing pixel-wise full-image prediction. However, to employ FCNs for HSIC, there are still two common problems that need to be solved, which are sparse labeling and memory overflow. As a traditional semantic segmentation method, FCNs require a standard semantic label map covered by sufficient pixel labels for the training loss calculation. Since HSI training labels are usually obtained by random sampling of the full label map, these labeled samples are sparsely distributed randomly in space and cause label vacancy (as shown in the green dashed box of Fig. 1). The absence of such spatially supervised information can lead to inadequate learning of network weights (as shown by the gray dashed line in the blue box of Fig. 1), which in severe cases may lead to failure converge or overfitting. In addition, as the pooling operations are rarely used in FCNs, many large-size feature maps are generated during the forward propagation, which occupies a lot of memory. Therefore, most of the current FCNs are not equipment-friendly methods.

To this end, this article proposes a novel lightweight hyperspectral FCN, namely, successive spatial rectified network (SSRNet). To ensure the lightweight scale, SSRNet is designed by splitting the complex deep network into several shallow subnetworks to reduce the large number of high-dimensional operations caused by the deepening of the network and then nesting the subnetworks for successive rectified learning to improve the classification performance of the model. It needs to be noted that SSRNet contains two types of shallow subnetworks: one is the local channel attention-based spectral FCN (LCA-SFCN) and the other is the separable atrous spatial pyramid module (SASPM). The former is designed with an LCA mechanism for input images of arbitrary size to improve the spectral classification accuracy. The latter takes the classification map of the previous module as input and performs the extraction of spatial feature on multiple receptive fields by constructing the separable atrous spatial pyramid. Furthermore, multiple SASPMs are concatenated and then attached to SSRNet for progressively rectifying the classification map. Besides, to reduce the MAC, SSRNet is designed as a patch-wise FCN, which uses the small-size patch samples as input rather than full hyperspectral images during training, yet predicts image with arbitrary size. To construct the patch-wise pixel-to-pixel training dataset and mitigate the impact of sparse labels, a novel noncentral positional sampling (NCPS) strategy is also proposed, which enables each labeled pixel to equally traverse all spatial positions of each training patch through the shift-sampling operation, so as to achieve the data augmentation. Meanwhile, NCPS supports the spatial uniform coverage of the semantic label maps by each labeled pixel in a statistical sense and the fair usage of different labeled pixels at fixed locations of the semantic label map.

The main contributions of this article are summarized as follows.

1) In this article, a lightweight FCN is proposed with the name of SSRNet, which structurally consists of several shallow subnetworks with depth-wise separable convolutions. In addition, due to the use of small patches instead of full hyperspectral images for pixel-to-pixel training, SSRNet can significantly reduce the training memory overhead while maintaining the fast inference ability.
2) This article proposes an LCA mechanism, which can improve the classification accuracy of spectral FCN. By adequate localizing operations, LCA can prevent the computation of channel attention from being affected by the change in image size.
3) This article also introduces a separable atrous convolution (SAC), which combines atrous convolution and depth-wise separable convolution operations. Then, a lightweight SASPM is proposed to perform multireceptive field representation and extract spatial features in high resolution.
4) To construct the patch-wise semantic training set, an NCPS strategy is proposed, which enables sparse labeled pixels to uniformly traverse all spatial locations on the label map by shift sampling. In a statistical sense, this fills the spatial vacancy of semantic labels and alleviates the sparse label problem of hyperspectral pixel-to-pixel training.

The remainder of this article is organized as follows. Section II summarizes the existing work and technical principles. Section III describes the proposed method and the specific structure of SSRNet in detail. The comparison experiments and classification results are presented in Section IV. Then, Section V presents the ablation experiments and performance analysis of SSRNet. Finally, the conclusion is given in Section VI.

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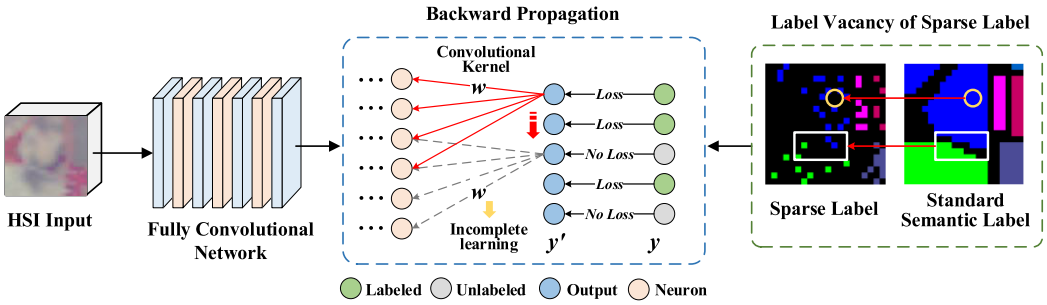


Fig. 1. Effect of sparse labels in terms of FCN training.

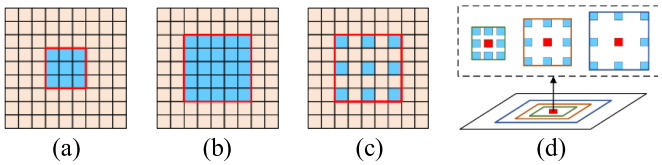


Fig. 2. (a) Standard convolution ($k = 3$). (b) Standard convolution ($k = 5$). (c) Atrous convolution ($k = 3$ and $d = 2$). (d) ASPP.

II. Related Works

A. Fully Convolutional Networks for HSI Classification

Due to the limitation of fully connected layers, traditional CNN classification networks can only deal with the scene classification task with fixed-size images. To handle the pixel-level density prediction task, Long et al. [25] proposed the FCN, which replaces the fully connected layer with convolutional operations and adopts an encoding–decoding structure with skip connections. It opens the way for highly accurate pixel-level classification based on CNN. Subsequently, Unet [26], SegNet [27], RefineNet [28], and many classical networks containing attention mechanisms [29], [30], [31] have been proposed one after another. Among them, some networks, such as DeepLab [32], have adopted the atrous/dilated convolution for achieving the fine-grained segmentation. As shown in Fig. 2(a)–(c), atrous convolution is able to control the size of receptive field by adding voids to the standard convolution kernel. The equivalent convolution kernel size of the atrous convolution can be calculated by $k + 2 \times (d - 1)$, where k is the kernel size and d is the dilation rate to control the receptive field. In practice, the atrous spatial pyramid pooling (ASPP) structure is generally employed by combining atrous convolutions with different dilation rates for multiscale target prediction, as shown in Fig. 2(d). Recently, studies on the FCNs application to HSIC are mainly concentrated in recent years. Initially, FCNs are used only as feature extraction method. For example, pretrained FCN and DeepLab are, respectively, employed to extract effective discriminative features followed by traditional classification methods such as SVM [33], [34].

To our best knowledge, Xu et al. [35] designed a dual-branch FCN, which is the first implementation of pixel-to-pixel training pattern in an HSIC task. The spectral branch consists of only a few simple 1×1 convolutional layers, while the spatial branch consists of 3×3 convolutions with residual connections. It is proved that the pixel-to-pixel pattern enables FCNs much faster than the traditional patch-based networks. Since FCNs take the whole hyperspectral image as input and are trained in a single-sample iterative manner, such networks are classified as image-based methods in this article. In addition, Pan et al. [36] proposed an FCN based on atrous convolution, and it is demonstrated that atrous convolutions with multiple dilation rates can somehow improve the resolution of feature map and achieve fine-grained classification. Furthermore, a criss-cross nonlocal module is introduced into the FCN to capture long-range context information for enhancing the position response of pixel-level classification [37]. However, although FCNs are competent for fast classification of hyperspectral images, these models suffer from the problems of massive memory cost and heavy computation consumption. Especially for compact devices, excessive MAC can even cause compiling obstacles.

B. Lightweight Convolution for HSI Classification

As the CNNs deepen, a single convolution operation often involves the simultaneous computation of hundreds or thousands of feature maps. Especially for hyperspectral data with hundreds of bands, such convolution operation over vast channels can generate a very considerable number of parameters and computational cost. Therefore, an effective method to lighten CNN networks is to use factorized convolution [38], which splits the standard convolution into two or more lightweight convolutions while keeping the size of the receptive field constant. As one of the infusive factorized convolution methods, depth-wise separable convolution decomposes standard convolution into a depth-wise convolution and a point-wise convolution. The depth-wise convolution carries out the spatial filtering for each input channel with a single convolution kernel, followed immediately by a 1×1 point-wise convolution, to control the output channels. To compare the difference in parameters and floating-point operations (FLOPs) between depth-wise separable convolution and standard convolution, it is assumed that for an input feature map with the size of $H \times W \times C_{\text{in}}$; if C_{out} convolution kernels with size of $k \times k$ and an appropriate padding are given, the output feature map size after convolution is $H \times W \times C_{\text{out}}$. Thus, the theoretical parameter compression ratio r_c and the computation speedup ratio r_s are calculated as follows:

$$r_c = \frac{\text{Param}_{\text{DSC}}}{\text{Param}_{\text{SC}}} = \frac{k \times k \times C_{\text{in}} + C_{\text{in}} \times C_{\text{out}}}{k \times k \times C_{\text{in}} \times C_{\text{out}}} = \frac{1}{C_{\text{out}}} + \frac{1}{k^2} \tag{1}$$

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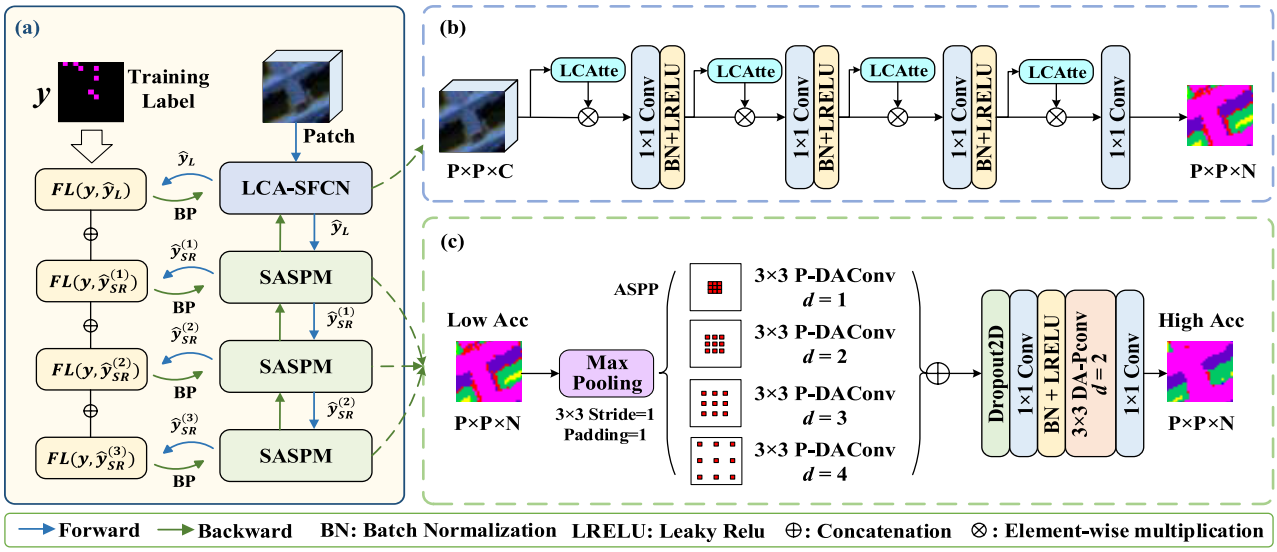


Fig. 3. Overview of the successive spatial rectified network. The SSRNet consists of LCA-SFCN and three SASPMs. (a) Successive spatial rectified learning. (b) LCA-SFCN: LCA spectral-FCN. (c) SASPM.

$$r_s = \frac{\text{FLOPs}_{\text{DSC}}}{\text{FLOPs}_{\text{SC}}} = \frac{k \times k \times C_{\text{in}} \times H \times W + C_{\text{in}} \times C_{\text{out}} \times H \times W}{k \times k \times C_{\text{in}} \times C_{\text{out}} \times H \times W} = \frac{1}{C_{\text{out}}} + \frac{1}{k^2}. \quad (2)$$

Moreover, some other studies have shown that compared to the significantly decreasing amplitude of parameters and FLOPs, there is only a little performance loss caused by depth-wise separable convolution [39], [40]. Therefore, depth-separable convolution is widely employed in the design of lightweight structure network.

III. Methodology

A. SSRNet for HSI Classification

To achieve fast and high-precision classification of hyperspectral images on a small platform, this article proposes a low-memory-dependent lightweight hyperspectral FCN, i.e., SSRNet. SSRNet consists of an LCA-SFCN and several SASPMs connected in series, whose structure is shown in Fig. 3. SSRNet first completes the initial classification of hyperspectral images by LCA-SFCN to produce a feature classification map with the same size as the input. Then, a successive spatial rectified structure is formed by attaching some SASPMs sequentially, via which the coarse-gained map is progressively rectified to output the fine-grained result. By the way, a structural parameter S is introduced to control the number of SASPMs, that is, the number of performed spatial corrections. To balance the lightweight and accuracy, the structural parameter S is set to 3 according to the experimental results in Section V-A. In addition, each subnetwork of SSRNet produces the prediction map to calculate the loss corresponding to the semantic label. It is worth noting that the unlabeled position information in the patch needs to be ignored when calculating the loss using sparse semantic labels.

B. Successive Spatial Rectified Learning

With constant convolutional kernel size and feature map dimensions, the number of parameters and FLOPs of a single convolutional layer depends only upon the input and output channels. To keep competitive representation ability, the number of feature map channels increases as the network deepens. Thus, most of the parameters and computational effort in a CNN come mainly from the high-dimensional structure in the tail of the network. In view of this, a successive spatial rectified learning method is proposed so as to further reduce the network parameters and computational cost. The main idea of this method is to split the complex deep network into several shallow subnetworks to avoid the continuous increase of feature channels throughout the whole network, while multiple loss functions corresponding to different depths are designed to guarantee that the subsequent rectified module can gradually fine-tune the classification map of the previous module to improve the accuracy. In these shallow subnetworks, LCA-SFCN aims to reduce the dimension of the original data and achieve the categories coding, i.e., $f_L: X^{P\times P\times C} \rightarrow Y^{P\times P\times N}$, where C is the number of bands, N is the number of categories, and C is much larger than N. In the encoding process, LCA-SFCN compresses the spectral information of features and completes the preliminary classification. Since it is not probabilistically processed by *Softmax*, this preliminary classification map can be regarded as the feature map with enriched information and also used as the input of subsequent SASPMs. Although each SASPM [i.e., $f_{\text{SR}}^{(i)}(\cdot)$] has a limited representation capability due to their shallow structure, SSRNet can be gradually fit into a high-precision classification network [i.e., $f_H(\cdot)$] by successively nesting the SASPMs several times, as shown in the following:

$$f_H(x) \Rightarrow f_{\text{SR}}^{(n)}\left(f_{\text{SR}}^{(n-1)}\left(\cdots\left(f_{\text{SR}}^{(1)}(f_L(x))\right)\cdots\right)\right). \quad (3)$$

In the case of attaching multiple SASPM modules, the classification of the subsequent module relies heavily on the

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output of the previous module. Due to the drastic fluctuation of training parameters, the module is unable to produce a more stable classification map quickly, resulting in unstable training of the next module and a chain reaction affecting the overall training of the whole network. In this regard, SSRNet designs the total loss function as the sum of the losses of each module. By calculating the losses separately at different depths, the module close to the input can perform more parameter updates to accelerate convergence, thus providing more stable classification maps for subsequent modules.

However, as the number of corrections increases, the input classification map already has a high accuracy itself before being sent to SASPM. At this moment, the module often has difficulty to improve the accuracy or even directly degrades to an identity mapping because of the tiny loss. To cope with this problem, a focal loss [41] is introduced in SSRNet, which is a loss function that can adjust the calculated loss value according to the classification difficulty of the training samples, and its formulas are shown as follows:

$$p_t = \begin{cases} \hat{y}, & y = 0 \\ 1 - \hat{y}, & \text{otherwise} \end{cases} \tag{4}$$

$$\mathrm{FL}(y, \hat{y}) = -\frac{(1 - p_t)^{\gamma}}{n} \sum \log(p_t) \tag{5}$$

where n is the number of labeled samples in the same batch and $\hat{y}$ and y are the prediction results and one-hot encoding of the ground truth, respectively. Compared to the cross-entropy loss, the focal loss has increased a modulating factor $(1 - p_t)^{\gamma}$, where γ is set to 2 in this study. For easy samples, the modulating factor is close to 0, to decline their contributions for loss. On the contrary, hard samples can make the modulating factor approach 1 to retain more loss, which ensures a more reasonable training process.

C. Spectral Fully Convolutional Network With Local Channel Attention Mechanism

Although the accuracy of SSRNet can be gradually improved by spatially rectified learning strategies, it is still very essential to optimize the initial spectral classifier for the best possible performance. A common improvement method is to add the channel attention module to the spectral classification network [42]. Traditionally, the channel attention is calculated by adaptive pooling operations, which compress the raw data into two vectors by max pooling and average pooling. Then, they are further compressed by r times in the channel dimension through a linear layer. After that, these two vectors will be restored to their original size and summed together. Finally, the attention weights are obtained by a sigmoid function. The SSRNet can only complete channel attention learning based on the local patch of fixed size during the training stage. However, the input size of the network may not be the same during the inference stage, which can lead to the failure attention learning from previous training. In response, this article proposes an LCA method applicable to situations where the input patch size is significantly changeable between the training and inference stages. The main idea of LCA is to locally divide the large image into local unit of the same size as the training patches during the inference stage and to complete the attention calculation within each local unit. As shown in Fig. 4, to achieve local division, the adaptive pooling operation is abandoned in LCA and window pooling is used instead, whose window size and stride are the same as the patch size in the training stage. Furthermore, the LCA employs a 1×1 convolution operation instead of linear layers for attention calculation in the 2-D case, to accomplish attention calculation, and finally, the attention weights of the entire image are generated based on the nearest neighbor interpolation method. The structure of LCA-SFCN in SSRNet is shown in Fig. 3(b). The spectral FCN consists of four 1×1 convolutional layers, and the number of kernels in all convolutional layers is 69 except the last layer. Before each convolution layer, the LCA module will be carried out upon the feature map first, while attention information will be added to the features by element multiplication. In this article, the channel compression ratio r of LCA is set to 3.

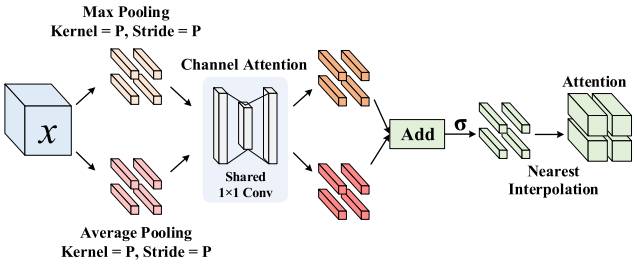


Fig. 4. Structure of LCA mechanism.

D. Separable Atrous Spatial Pyramid Module for Spatial Rectification

SASPM takes the low-precision preliminary classification map as the input and outputs the higher precision classification map by spatial rectification. To suppress the frequent pepper noise in the low-precision classification map, SASPM first smoothens the input image by maximum pooling, as shown in Fig. 3(c). It should be noted that a 3×3 kernel with one stride and proper padding operation are performed for keeping the image size unchanged. Then, the feature maps are fed into the ASPP module for multiscale feature extraction, which is composed of four atrous convolutions with different dilation rates. Meanwhile, to ensure the network as lightweight as possible, an SAC is constructed in this study to replace the standard atrous convolution. The SAC advances the depth-wise convolution of depth-wise separable convolution into depth-wise atrous convolution, which adopts dilated kernels in depth-wise convolutional operations. Furthermore, as shown in Fig. 5(b)–(c), two different forms of SAC, i.e., "DA-P" and "P-DA" convolution, are designed to meet different requests. Similar to the depth-wise separable convolution, the "DA-P" convolution performs depth-wise atrous convolution before point-wise convolution, while the "P-DA" convolution performs 1×1 point-wise convolution before depth-wise atrous convolution. As for ASPP module, "P-DA" is introduced to ascend dimension space and avoid the spatial representation bottleneck by employing point-wise

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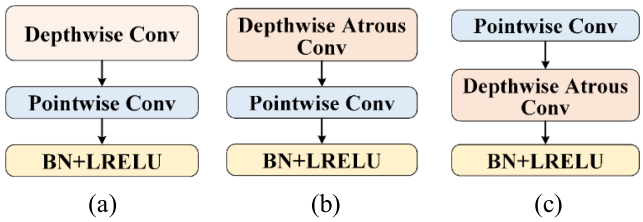


Fig. 5. (a) Depthwise separable convolution. (b) "DA-P" convolution. (c) "P-DA" convolution.

convolution first. The four "P-DA" convolution branches are employed to extract the spatial rectified features with different receptive fields. Then these features are concatenated together and the random deactivations are performed upon them by 2-D dropout to improve the generalization performance of SSRNet. After that, the 1×1 convolution is used to compress the feature maps. Considering that the compressed features are already information-rich, they can be further fused directly by "DA-P" convolution without any dimension ascending operation. Finally, the ultimate classification result of SSRNet will be produced by another 1×1 convolution.

E. Noncentral Positional Sampling Strategy for Hyperspectral Sparse Labels

Typically, to support pixel-to-pixel training, FCNs need to use semantic label maps with high label coverage, which is the percentage of labeled pixels in the label map among all pixels. However, for hyperspectral images, the sparse distribution of pixel labels results in a lack of available labels at most positions of training patch, which leads to an uneven distribution of sample labels at different positions. In this way, the network relies on only a small number of sparse labels for loss calculation and backpropagation, which leads to insufficient training. To solve the above problems, a novel NCPS strategy is proposed in this article. The main idea of NCPS is to make every pixel label appear at any position on the training patch by shift sampling, for the purpose of compensating for the spatial deficit caused by sparse labels and ensuring adequate training as well as quick convergence.

The procedure of NCPS is shown in Fig. 6. First, sparsely distributed training labels are generated by random sampling. Then, NCPS generates the corresponding sample groups by continuously shifting the patch window and conducting patch-wise sampling at the same time, in which each training semantic label is equally reused in each position within the whole patch. In this case, the offset labels in other sample groups can statistically compensate for the missing information of the semantic label graph. Moreover, for each labeled pixel, the shift-sampling groups can provide a diversified positional description, thus reducing the dependence of the network on the fixed spatial distribution pattern and improving the discrimination ability of the network. However, there are duplicates and redundancies in the expanded samples resulting from such pixel-by-pixel shift sampling. Therefore, NCPS needs to successively perform intragroup filtering and intergroup merging operations to achieve a refined and simplified processing of these samples. If the size of training patch is P, then each sample group includes P^2 offset samples. To reduce redundancy, each sample group is then screened for intragroup samples. Given the uniformity of label position distribution, a random sampling method is adopted to perform the intragroup screening, and the percentage of retained offset samples in each group is determined by the sampling coverage rate $\alpha \in [1, 0]$. Since some labeled pixels are close to each other, the duplicate may occur in different sample groups after shift sampling. Thus, intergroup merging is also required after intragroup selection to ensure that only one of the duplicates can be retained among different sampling groups. By setting an appropriate sampling coverage and performing a unique merging operation, the final number of samples by NCPS will be controlled within an acceptable range. The effects of different sampling coverage rates on sample number and network performance are discussed in Section V-B.

IV. Experimental Results

To verify the validity of the proposed method, experiments have been conducted on three benchmark hyperspectral datasets, including Indian Pines (IP), University of Pavia (UP), and University of Houston (UH) datasets.

A. HSI Datasets

The IP dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over the IP agricultural test site in northwestern Indiana in 1992. It includes 224 bands in the wavelength range from 0.4 to 2.5 μm, with 200 spectral bands retained after eliminating 24 bands absorbed by water. This dataset is in the size of 145×145 pixels with a spatial resolution of 20 m, and the feature labels contain 16 different feature categories.

The UP dataset was gathered by the Reflective Optics System Imaging Spectrometer (ROSIS) in 2003 at the campus of Pavia University in northern Italy. It contains 115 spectral bands in the wavelength range from 0.43 to 0.86 μm, with a spatial resolution of 1.3 m. This dataset consists of 610×340 pixels, with 103 bands per pixel after removing 12 bands that are significantly affected by noise. There are nine different categories of land covers in the UP scene.

The UH dataset was collected by the ITRES CASI-1500 sensor over the UH campus and the surrounding urban area, which is also used in a data fusion competition organized by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS2013). This dataset consists of 349×1905 pixels and 144 spectral bands in the wavelength range from 0.38 to 1.05 μm, with 15 categories labeled.

B. Experimental Settings

To standardize the data size, all hyperspectral datasets are preprocessed by the min–max normalization. After that, each category in the three datasets is divided into the training and validation sets with the sample size of 150 and 50, respectively, and the other remaining samples are used as test sets. In particular, the categories with less than 200 samples will be divided in a ratio of 1:1:2, and the specific sample divisions on three datasets are shown in Tables I–III. Besides,

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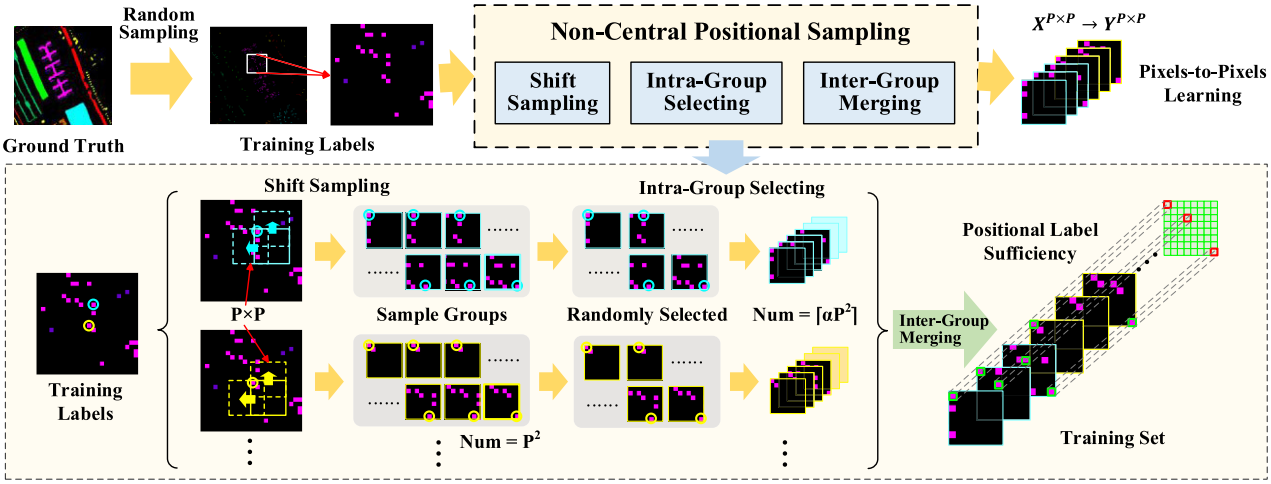


Fig. 6. Flowchart of NCPS strategy.

TABLE I

CATEGORY AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE IP DATASET

No.	Class	Train.	Val.	Test.
1	Alfalfa	12	11	23
2	Corn-notill	150	50	1228
3	Corn-mintill	150	50	630
4	Corn	150	50	37
5	Grass-pasture	150	50	283
6	Grass-trees	150	50	530
7	Grass-pasture-mowed	7	7	14
8	Hay-windrowed	150	50	278
9	Oats	5	5	10
10	Soybean-notill	150	50	772
11	Soybean-mintill	150	50	2255
12	Soybean-clean	150	50	393
13	Wheat	150	50	5
14	Woods	150	50	1065
15	Buildings-Grass-Trees-Drives	150	50	186
16	Stone-Steel-Towers	23	23	47
Total	-	1847	646	7756

TABLE II

CATEGORY AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE UP DATASET

No.	Class	Train.	Val.	Test.
1	Asphalt	150	50	6431
2	Meadows	150	50	18449
3	Gravel	150	50	1899
4	Trees	150	50	2864
5	Painted Metal Sheets	150	50	1145
6	Bare Soil	150	50	4829
7	Bitumen	150	50	1130
8	Self-Blocking Bricks	150	50	3482
9	Shadows	150	50	747
Total	-	1350	450	40976

TABLE III

CATEGORY AND THE NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE UH DATASET

No.	Class	Train.	Val.	Test.
1	Healthy grass	150	50	1051
2	Stressed grass	150	50	1054
3	Synthetic grass	150	50	497
4	Trees	150	50	1044
5	Soil	150	50	1042
6	Water	150	50	125
7	Residential	150	50	1068
8	Commercial	150	50	1044
9	Road	150	50	1052
10	Highway	150	50	1027
11	Railway	150	50	1035
12	Parking Lot1	150	50	1033
13	Parking Lot2	150	50	269
14	Tennis Court	150	50	228
15	Runing Track	150	50	460
Total	-	2250	750	12029

the training samples are randomly selected before each experiment, and all experiments are repeated five times with the average and variance retained. Finally, the overall accuracy (OA), average accuracy (AA), kappa coefficient, and the recall of each category are employed as the accuracy indicators for evaluating the classification performance.

For SSRNet, the hyperparameter S of network structure is set to 3, the dropout rate is set to 0.2, the activation function is selected as *LeakyReLU*, and the network loss function is devised as focal loss. In addition, the batch normalization technique is also widely adopted in SSRNet to alleviate the problem of gradient disappearance. The specific structure of SSRNet is shown in Table IV. SSRNet is trained using the samples obtained by the NCPS strategy. Meanwhile, the sampling coverage ratio α is set to 0.1, 0.1, and 0.2 for the datasets of IP, UP, and UH, respectively. Also, the training patch sizes are set as 13, 15, and 15. The Adam optimizer [43] is adopted in training with the weight decay of $1e^{-5}$. The initial learning rates are set to 0.01, 0.02, and 0.04, and the batch size is set to 256, 512, and 1024 for these three datasets. Besides, the "Poly" strategy [32] is applied for the learning rate decay, i.e., the learning rate is multiplied by $(1 - (\text{iter}/\text{max_iter}))^{0.9}$ for each iteration. Due to the design of multiple loss functions in stage of successive spatial rectified learning, SSRNet only

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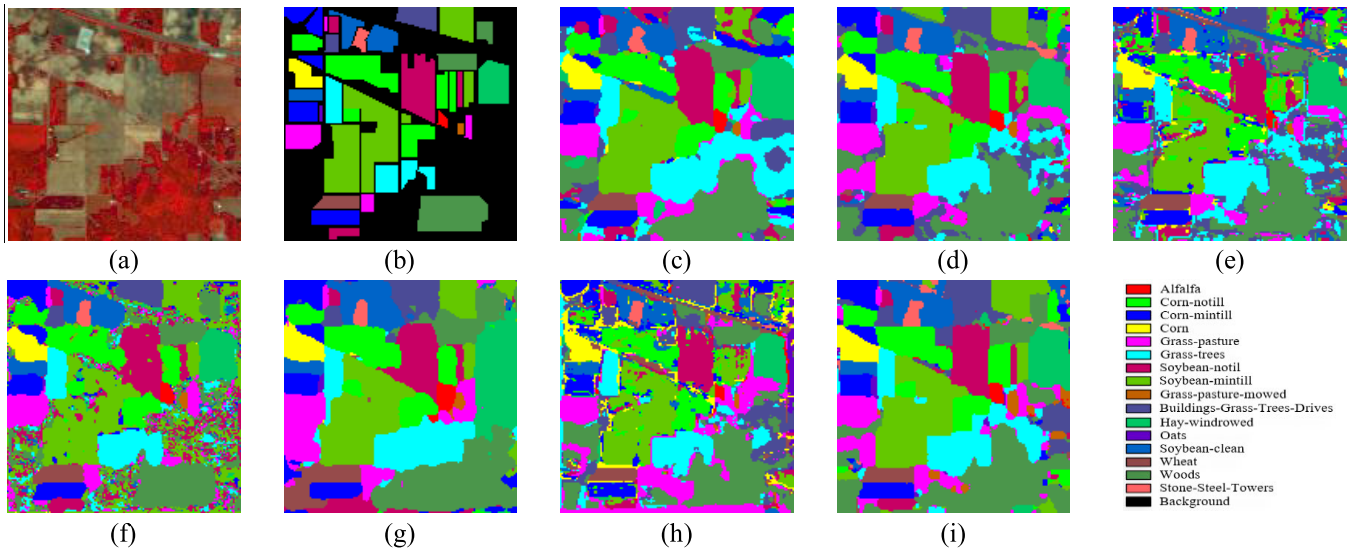


Fig. 7. Classification maps on the IP dataset. (a) False-color image. (b) Ground truth. (c) HybridSN. (d) A2S2KResNet. (e) SpectralFormer. (f) Unet-PCA. (g) FullyContNet. (h) SSFCN-CRF. (i) SSRNet.

TABLE IV
DETAILED STRUCTURE OF SSRNET ($S = 3$) WHILE LEAVING OUT BATCH NORMALIZATION OPERATIONS AND ACTIVATION FUNCTIONS

Module		Structure Details
LCA-SFCN	Local Channel Attention	$P\times P$ Pooling, stride = P, $r = 3$
	Conv_layer	1×1 Conv, 69
	Local Channel Attention	$P\times P$ Pooling, stride = P, $r = 3$
	Conv_layer	1×1 Conv, 69
	Local Channel Attention	$P\times P$ Pooling, stride = P, $r = 3$
	Conv_layer	1×1 Conv, 69
	Local Channel Attention	$P\times P$ Pooling, stride = P, $r = 3$
	Output layer	1×1 Conv, N
SASPM×3	Maxpooling	3×3 Pooling, padding = 1, stride = 1
	P-DA Conv_1	1×1 Point Conv, N×4
		3×3 Depth Atrous Conv N×4, $d = 1$
	P-DA Conv_2	1×1 Point Conv, N×4
		3×3 Depth Atrous Conv N×4, $d = 2$
	P-DA Conv_3	1×1 Point Conv, N×4
		3×3 Depth Atrous Conv N×4, $d = 3$
	P-DA Conv_4	1×1 Point Conv, N×4
		3×3 Depth Atrous Conv N×4, $d = 4$
	Dropout2D	Rate = 0.2
	Conv_layer	1×1 Conv, N×4
	DA-P Conv	3×3 Depth Atrous Conv N×4, $d = 2$
		1×1 Point Conv, N×4
	Ouput layer	1×1 Conv, N

needs a small number of iterations to converge. Therefore, the iteration number upon the IP dataset is set to just 80 in the experiments, and the other two datasets are both set to 60. In addition, the configuration of the experimental platform used in this article is I7-7700K 4.20 GHz, GTX1080Ti, and 16-GB RAM.

C. Classification Results

To fully validate the classification performance of SSRNet, every three SOTA methods are selected from the existing patch- and image-based classification methods as the comparison experiments. The selected patch-based methods include HybridSN [17], A2S2KResNet [22], and SpectralFormer [44], among which SpectralFormer is a nonconvolutional hyperspectral classification network based on Transformer [45] structure. Meanwhile, Unet, SSFCN [35], and FullyContNet [46] are the selected image-based comparative networks. Among them, Unet is a common lightweight semantic segmentation network. It is worth noting that the hyperspectral data are preprocessed using the principal component analysis (PCA) to reduce the spectral dimensions down to 3. Also, SSFCN is a lightweight spectral–spatial FCN based on a few convolutional layers. FullyContNet is an up-to-date image-based method that enables high-precision HSI classification due to the introduction of a variety of attention mechanisms and ASPP structure.

To truly reflect the classification performance of different methods, all comparative experiments comply with the optimal model configuration provided in the original paper and are conducted under the same hardware platform. The quantitative experimental results are shown in Tables V–VII. Generally, the SSRNet proposed in this article achieves the optimal classification results for all three datasets. For the IP dataset, the OA, AA, and kappa of SSRNet reach 99.57%, 99.67%, and 0.9949, respectively, increasing by 0.43–18.59%, 0.17–16.27%, and 0.0058–0.2208 compared to other methods. It is worth noting that SSRNet gains 100% recall when there are only 12, seven, and five training samples for the "Alfalfa," "Grass style owned," and "Oats" categories, respectively. For the UP dataset, the SSRNet achieves 99.75% for OA, 99.88% for AA, and 0.9967 for kappa with only about 3% of all labeled samples, which is superior to all the other methods. Among the nine categories, SSRNet has the best performance for seven categories in terms of the recall metric. In addition, SSRNet attains 99.78% of OA on the UH dataset with 0.37–5.2% improvement over other methods, and 8 of 15 categories reach 100% of recall.

Figs. 7–9 show the classification maps of different methods on three datasets. It can be found that there are many misclassified pixels within the intact terrain region in the classification maps of SpectralFormer, Unet, and SSFCN, which indicates that the accuracy of these models needs to be improved. In addition, although FullyContNet, A2S2KResNet,

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

No.	Patch-based			Image-based			Proposed
	HybridSN	A2S2KResNet	SpectralFormer	Unet-PCA	FullyContNet	SSFCN-CRF	SSRNet
1	99.13±1.74	98.26±2.13	78.26±8.70	**100±0.0**	**100±0.0**	79.13±11.25	**100±0.0**
2	92.49±1.03	**99.07±0.23**	76.11±5.47	91.29±2.09	96.99±0.62	44.85±10.15	99.01±0.61
3	92.32±1.70	**99.81±0.19**	86.38±3.34	96.38±1.00	99.62±0.22	82.25±3.13	99.75±0.26
4	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**	94.05±5.20	**100±0.0**
5	99.65±0.45	99.86±0.28	96.75±1.42	**100±0.0**	99.79±0.42	94.28±2.75	99.93±0.14
6	97.43±2.39	99.92±0.15	97.55±1.61	99.77±0.18	**100±0.0**	97.40±1.93	99.85±0.30
7	90±10.69	98.57±2.86	84.29±10.50	94.29±8.33	**100±0.0**	55.71±7.82	**100±0.0**
8	**100±0.0**	**100±0.0**	99.86±0.29	**100±0.0**	**100±0.0**	99.06±1.43	**100±0.0**
9	50±8.94	**100±0.0**	66±13.56	62±19.39	74±8.0	58±22.80	**100±0.0**
10	96.06±1.36	**99.79±0.30**	88.89±4.21	98.60±0.88	99.74±0.22	76.97±10.70	99.66±0.33
11	90.06±2.88	98.01±0.41	81.50±4.98	90.01±3.34	98.98±0.18	85.18±2.12	**99.43±0.46**
12	99.08±0.47	99.75±0.16	89.97±2.56	**100±0.0**	99.640.26	80.31±2.59	99.64±0.26
13	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**	**100±0.0**
14	97.13±1.76	99.59±0.29	93.16±1.84	99.72±0.25	99.92±0.04	96.47±0.97	99.92±0.11
15	99.46±0.48	99.68±0.65	95.16±4.43	**100±0.0**	**100±0.0**	97.10±0.82	**100±0.0**
16	91.06±3.40	**99.57±0.85**	92.34±6.25	94.04±0.85	96.60±1.70	93.62±5.12	97.45±0.85
OA(%)	94.12±1.18	99.14±0.09	86.59±1.54	95.13±1.06	99.08±0.07	80.99±2.57	**99.57±0.11**
AA(%)	93.37±0.96	99.49±0.14	89.14±1.69	95.38±1.39	97.83±0.47	83.40±2.06	**99.67±0.07**
$\kappa \times 100$	93.09±1.38	98.98±0.10	84.29±1.78	94.28±1.24	98.91±0.09	77.41±3.15	**99.49±0.13**

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET

No.	Patch-based			Image-based			Proposed
	HybridSN	A2S2KResNet	SpectralFormer	Unet-PCA	FullyContNet	SSFCN-CRF	SSRNet
1	90.40±3.81	99.65±0.27	86.96±6.67	98.56±0.85	99.60±0.37	95.59±1.59	**99.88±0.14**
2	97.43±1.19	99.19±0.53	92.87±4.79	88.26±2.19	99.51±0.03	97.93±0.73	**99.58±0.30**
3	94.52±1.63	99.38±0.38	89.32±3.42	97.29±1.41	99.30±0.06	96.90±0.94	**100±0.0**
4	99.15±0.31	99.53±0.12	97.74±1.10	97.70±2.15	99.59±0.65	98.03±0.94	**99.67±0.36**
5	**100±0.0**	**100±0.0**	99.97±0.07	99.97±0.07	**100±0.0**	99.32±0.55	99.98±0.03
6	96.33±5.17	99.86±0.23	91.75±5.51	99.94±0.12	99.97± 0.04	99.32±0.83	**100±0.0**
7	99.29±0.81	99.98±0.04	97.54±0.94	**100±0.0**	**100±0.0**	97.60±0.42	**100±0.0**
8	98.06±0.56	99.67±0.27	87.36±4.41	96.38±1.42	99.26±0.20	97.93±1.37	**99.82±0.09**
9	99.63±0.68	99.89±0.16	99.68±0.14	**99.97±0.05**	99.79±0.82	99.82±0.14	99.95±0.07
OA(%)	96.40±1.32	99.47±0.25	91.97±3.20	93.88±0.89	99.59±0.09	97.75±0.27	**99.75±0.11**
AA(%)	97.20±1.01	99.68±0.08	93.69±2.15	97.56±0.10	99.67±0.13	98.05±0.22	**99.88±0.03**
$\kappa \times 100$	95.18±1.77	99.29±0.33	89.36±4.17	91.98±1.12	99.44±0.12	96.98±0.36	**99.67±0.15**

and HybridSN can achieve high classification accuracy, their classification maps can only ensure the integrity of the ground objects within the region but have classification defects at the region boundaries. Benefiting from the successive spatial rectification of SASPM, the classification maps of SSRNet not only show smoother overall performance but also have a clearer delineation of local feature boundaries.

V. ANALYSIS OF EFFECTIVENESS

To verify the effectiveness of each component of the proposed SSRNet and the NCPS sampling strategy, this section devotes to the ablation experiments, comparison experiments on sampling coverage parameter, and sensitivity experiments of the training sample number. Finally, a comparative analysis of network complexity and inference speed is also presented at the end. It should be noted that a simple spectral FCN (SFCN) without any attention mechanism and trained with sparse labeled samples is used as the baseline in experiments.

A. Ablation Study

To investigate the contributions of the LCA mechanism, the number of SASPMs, the NCPS sampling strategy, and the SAC structure to SSRNet, ablation experiments are conducted on the UP dataset and the experimental results are shown in Table VIII. The OA, AA, and kappa of baseline are only 80.99%, 85.27%, and 0.7532, respectively. These classification metrics are significantly improved by 12.14%, 7.85%, and 0.1433 when adding the LCA module to the baseline. To further compare the performance difference between LCA and other attention mechanisms, the baseline is equipped with the traditional channel attention mechanism (CAM), spatial attention mechanism (SAM), and convolution block attention mechanism, which blends the channel and spatial

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UH DATASET

No.	Patch-based			Image-based			Proposed
	HybridSN	A2S2KResNet	SpectralFormer	Unet-PCA	FullyContNet	SSFCN-CRF	SSRNet
1	99.30±0.46	99.92±0.58	98.31±0.71	**100±0.0**	99.83±0.26	95.03±2.65	99.52±0.49
2	99.37±0.56	99.47±0.26	98.84±0.59	99.70±0.28	99.72±0.24	92.31±6.74	**99.83±0.17**
3	**100±0.0**	**100±0.0**	99.84±0.15	**100±0.0**	**100±0.0**	99.36±0.30	**100±0.0**
4	98.77±0.87	99.96±0.05	99.67±0.27	98.79±1.80	98.30±0.52	95.54±3.91	**99.98±0.04**
5	99.96±0.08	99.98±0.04	99.85±0.18	99.90±0.12	**100±0.0**	98.87±0.52	**100±0.0**
6	**100±0.0**	**100±0.0**	99.68±0.39	**100±0.0**	**100±0.0**	96.48±2.01	**100±0.0**
7	97.72±0.79	99.42±0.70	92.75±2.63	97.13±0.43	99.31±0.32	93.60±3.10	**99.89±0.14**
8	92.51±1.68	97.47±1.07	91.46±0.41	93.72±2.33	94.12±0.90	90.98±2.48	**99.48±081**
9	97.81±1.77	99.26±0.89	90.70±2.36	97.74±1.30	97.51±1.19	91.84±2.48	**99.37±0.65**
10	99.16±0.48	99.69±0.35	90.81±2.45	99.57±0.86	**100±0.0**	95.011.67	**100±0.0**
11	97.86±0.76	99.88±0.28	93.18±1.48	99.52±0.39	**100±0.0**	93.08±3.14	**100±0.0**
12	96.94±0.73	98.14±0.98	94.91±1.25	96.19±1.93	98.84±0.67	95.53±1.66	**99.38±0.74**
13	97.70±2.38	**100±0.0**	93.53±1.67	**100±0.0**	**100±0.0**	91.67±2.95	**100±0.0**
14	**100±0.0**	**100±0.0**	99.82±0.35	**100±0.0**	**100±0.0**	98.86±0.67	**100±0.0**
15	**100±0.0**	**100±0.0**	99.96±0.09	**100±0.0**	**100±0.0**	97.70±1.05	**100±0.0**
OA(%)	98.16±0.21	99.41±0.14	95.54±0.53	98.46±0.18	98.92±0.13	94.58±0.90	**99.78±0.16**
AA(%)	98.47±0.21	99.55±0.11	96.22±0.50	98.82±0.14	99.17±0.10	95.06±0.74	**99.83±0.13**
$\kappa \times 100$	98.00±0.23	99.36±0.15	95.16±0.57	98.32±0.19	98.83±0.14	94.11±0.98	**99.76±0.18**

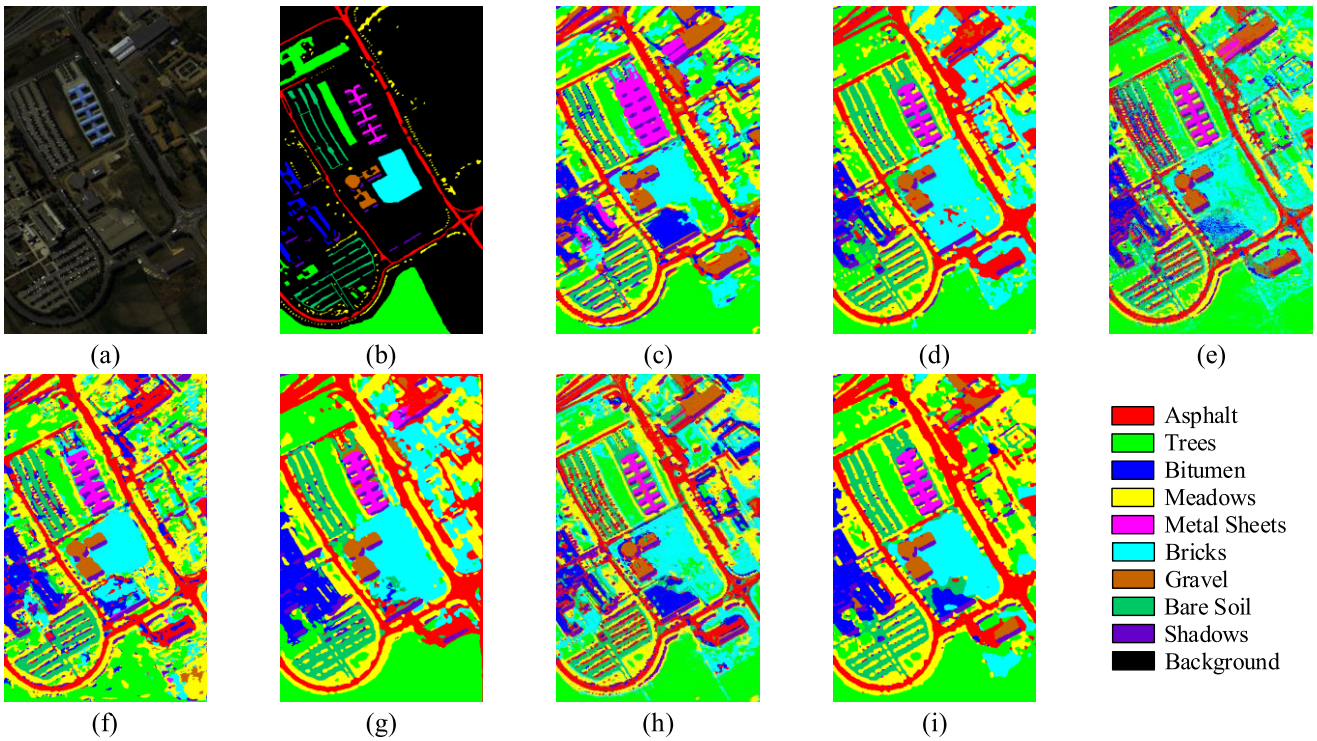


Fig. 8. Classification maps on UP dataset. (a) False-color image. (b) Ground truth. (c) HybridSN. (d) A2S2KResNet. (e) SpectralFormer. (f) Unet-PCA. (g) FullyContNet. (h) SSFCN-CRF. (i) SSRNet.

attention. The experimental results are shown in Fig. 10. It is seen that SAM has roughly no gain for SFCN which relies only on spectral information, while CAM or CBAM instead significantly reduces the accuracy of baseline. This can be attributed to that the adaptive pooling used in CAM and CBAM leads to the failure of attention calculation, as the input image size has been obviously changed from the training stage to the inference stage. On the contrary, the proposed LCA guarantees that the attention region does not change with the variations of input image size by localization operations, thereby successfully improving the OA of baseline to over 90%. Subsequently, the SASPM module is introduced on top of LCA, and the spatial rectification information brought by the SASPM further increases the OA by 5.25% compared to that of utilizing only spectral information. Previously, the network was trained with sparse semantic labels, and after

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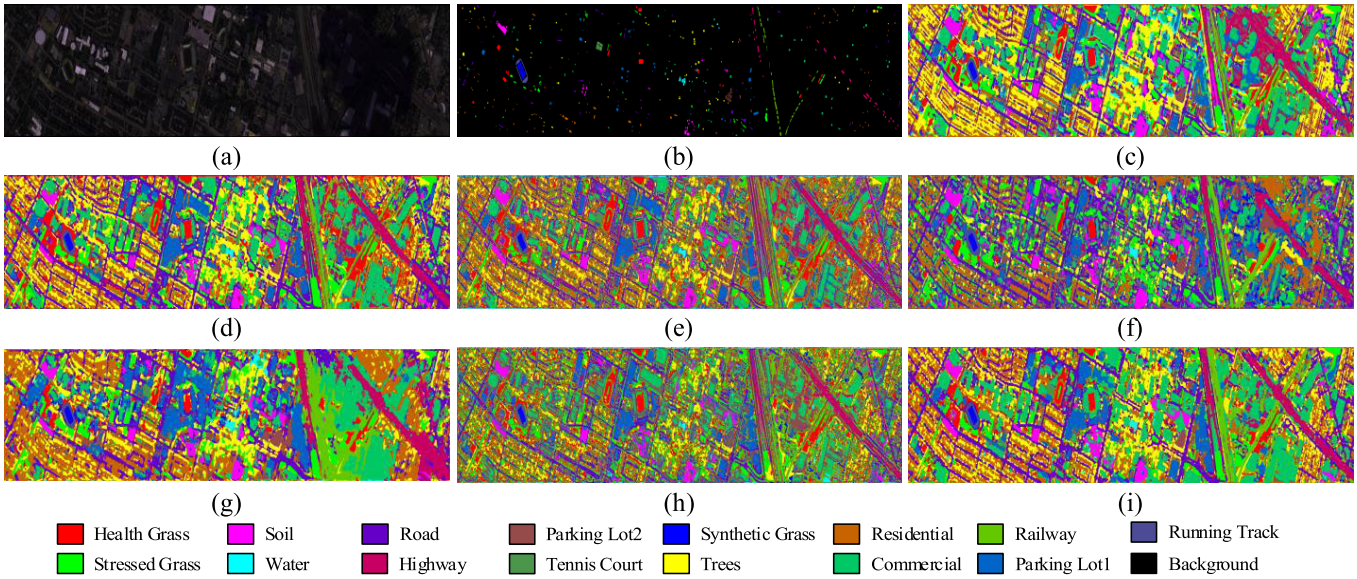


Fig. 9. Classification maps on the UH dataset. (a) False-color image. (b) Ground truth. (c) HybridSN. (d) A2S2KResNet. (e) SpectralFormer. (f) Unet-PCA. (g) FullyContNet. (h) SSFCN-CRF. (i) SSRNet.

TABLE VIII
CLASSIFICATION RESULTS AND RELATIVE PARAMETERS OF DIFFERENT ABLATION EXPERIMENTS ON THE UP DATASET

Method	LCA	SASPM	NCPS	OA(%)	AA(%)	Kappa	Params(K)
Baseline				80.99	85.27	0.7532	17.6
+LCA	√			92.13	94.12	0.8965	34.2
+LCA+SASPM	√	1		97.38	98.49	0.9651	44.3
SSRNet	√	1	√	99.42	99.59	0.9927	44.3
SSRNet	√	2	√	99.68	99.77	0.9957	54.5
SSRNet	√	3	√	99.75	**99.88**	0.9967	64.6
SSRNet	√	4	√	99.59	99.77	0.9945	74.8
SSRNet	√	5	√	99.45	99.54	0.9926	84.9
SSRNet *w/o* SAC	√	3	√	**99.83**	99.86	**0.9977**	122

SSRNet *w/o* SAC: SSRNet without separable atrous convolution

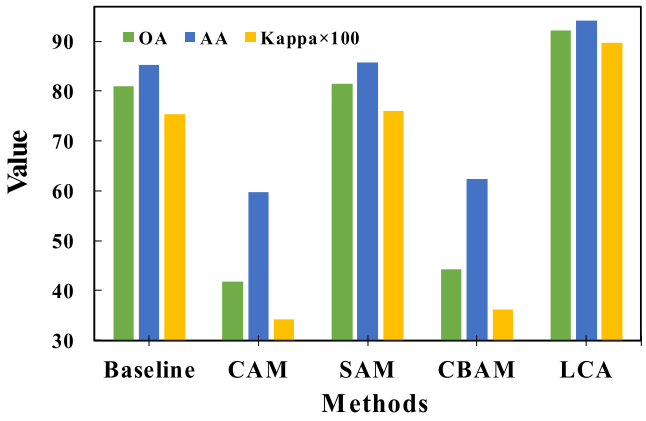


Fig. 10. Histogram of OA, AA, and kappa for SFCNs with different attention modules.

the introduction of NCPS, the network achieved the accuracy of over 99%. The column of SASPM means the number of SASPMs attached to SSRNet, which is the hyperparameter S. As the module increases, the accuracy also improves. It can be observed from Table VIII that there are only 64.6K parameters of SSRNet in total after three rectifications since each SASPM will only result in an approximate 10k parameters increase. However, after the number of SASPMs exceeded three, the classification performance of SSRNet tended to plateau. Therefore, S is set to 3 for the standard SSRNet in this study. Furthermore, as shown in the last row of Table VIII, if the SAC in SSRNet is replaced by the standard atrous convolution, the classification accuracy is only slightly improved, while the model parameters are approximately increased to double. This suggests that the SAC introduced into SSRNet can effectively reduce the model parameters while maintaining a comparable accuracy to that of the standard atrous convolution.

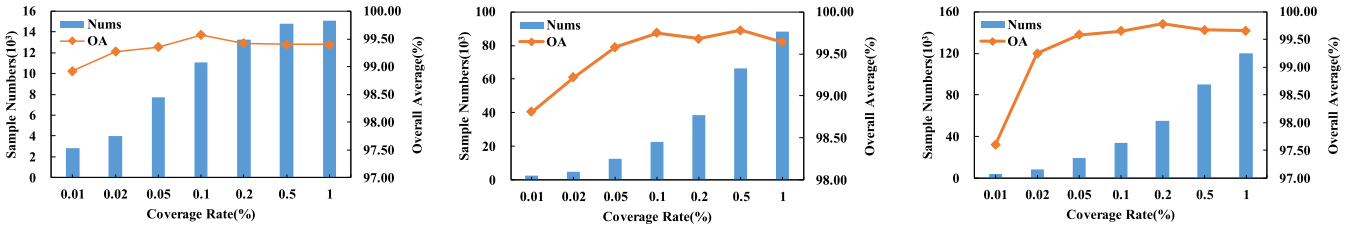
B. Impact of Sampling Coverage Rate

Fig. 11 shows the variation of the number of training samples obtained by NCPS with the OA of SSRNet at different sampling coverage rates α. Overall, with the increase of sampling coverage rate increases, the total training samples collected by NCPS also increase. However, as there is partial overlap among different sample groups obtained through shift sampling, the growth rate in the number of training samples decreases as α increases. The experimental results show that the NCPS can obtain up to 15k, 88k, and 120k samples, respectively, on IP, UP, and UH datasets ($\alpha = 1$), which is equivalent to an eight- to 60-fold expansion of the original labeled training samples. In addition, the OA of SSRNet gradually increases with the growing of coverage rate α and the total number of samples. However, the accuracy will soon reach saturation when the sampling coverage rate increases to a certain level. If the coverage rate continues to increase, the overall classification accuracy will fluctuate significantly. This is mainly because NCPS adopts dense shift sampling with a stride of 1, which leads to the generation of more and more similar samples as the coverage rate increases. These redundant samples not only slow down the speed of training but also affect the accuracy. Therefore, in practical applications, it is necessary to determine the appropriate sampling

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TABLE IX
CONTRAST IN TERMS OF PARAMETERS, FLOPS, AND MAC AMONG DIFFERENT METHODS ON THREE DATASETS

Methods		IP			UP			UH		
		Params(M)	FLOPs(G)	MAC(MB)	Params(M)	FLOPs(G)	MAC(MB)	Params(M)	FLOPs(G)	MAC(MB)
Patch-based	HybridSN	5.122	0.496	3.44	4.845	0.102	29.99	4.846	0.102	30.00
	A2S2KResNet	0.371	0.202	13.84	0.221	0.258	16.56	0.283	0.358	23.02
	SpectralFormer	0.343	-	12.98	0.177	-	11.86	0.226	-	9.27
Image-based	Unet	34.528	41.50	343.96	34.528	411.46	2168.89	34.527	1314.52	6669.10
	FullyContNet	1.371	17.90	169.24	1.314	155.10	1554.53	1.338	527.40	5075.14
	SSFCN	0.109	14.24	84.54	0.096	120.77	737.90	0.102	409.66	2505.54
Proposed	SSRNet ($S = 1$)	**0.090**	**0.018**	**2.57**	**0.044**	**0.012**	**2.15**	**0.070**	**0.020**	**3.05**
	SSRNet ($S = 2$)	0.119	0.028	4.18	0.054	0.016	3.31	0.096	0.032	5.02
	SSRNet ($S = 3$)	0.149	0.038	5.79	0.065	0.020	4.47	0.122	0.044	6.98



(a) (b) (c)

Fig. 11. Overall accuracy and sample number of NCPS training set with different sampling coverage rates on three datasets. (a) IP dataset. (b) UP dataset. (c) UH dataset.

coverage rate according to the total number of training samples and classification accuracy. Based on the above experimental results, the coverage rates of 0.1, 0.1, and 0.2 are chosen for these three datasets, respectively, resulting in 11k, 23k, and 34k samples for training.

C. Sensibility Analysis of Training Samples

To investigate the sensitivity of SSRNet to the number of training samples, comparison experiments are conducted by using different numbers of training samples of 10, 50, 100, 150, and 200 per class. In addition to the six comparison methods mentioned above, the experimental methods also include SSRNet adopting different sampling strategies, such as the SSRNet-SLS that directly uses sparsely labeled samples, the SSRNet-Image that uses an image-based full-image training strategy, and the SSRNet-NCPS that uses the NCPS strategy. As shown in Fig. 12, the sensitivity of different methods to the number of training samples is examined on three datasets by means of the OA. The proposed SSRNet-NCPS method outperforms the other methods for all cases of training samples, especially for small samples. More specifically, when there are only 100 samples for each category, the OA of SSRNet-NCPS has exceeded 99% on the three datasets and also over 95% even if each category has only 50 samples. This can be attributed that NCPS implements a data augmentation of the sample, and its advantage becomes more evident in the case of fewer training samples available. With only ten training samples reserved for each category, the actual number of unique labeled samples that can be provided by IP, UP, and UH datasets is only 154, 90, and 150, while NCPS can expand these samples into quantities of 2080, 1836, and 6244, respectively. These pixel-level semantic labels, which are uniformly distributed at different positions in the patches, can effectively avoid the spatial information loss in sparse situations. Therefore, even when only ten label pixels are allocated to each category, the proposed SSRNet-NCPS method still achieves 81.11%, 92.03%, and 86.21% of OA on the benchmark datasets, which is more than 15% higher compared to that of SSRNet-SLS and 7.9–9.4% higher compared to that of SSRNet-Image. These experimental results demonstrate that even in the case of small samples, the NCPS strategy can effectively guarantee the high-precision classification performance of the SSRNet.

D. Complexity and Speed Analysis

Table IX presents the model parameters, FLOPs, and MAC of SSRNet versus other neural networks on the three datasets. The SSRNet ($S = 3$) contains three SASPMs, whose parameter number ranges from 0.065 to 0.149 M, which relies on the number of categories, thus slightly higher than the shallow network SSFCN in terms of IP and UH datasets but lower than other networks. By cutting down one or two of the spatial correction modules, the number of network parameters can be lowered to less than 0.1 M, while SSRNet can still maintain a competitive classification accuracy. In addition, it can also be seen from Table IX that image-based methods usually have much higher FLOPs and MAC than patch-based methods because image-based methods require the whole hyperspectral images to complete end-to-end training. As a patch-wise FCN, SSRNet not only inherits the low memory advantage from patch-based methods but also combines a lightweight structure of spatial rectified learning with depth-wise separable convolution technique. For a single forward computation, SSRNet only requires less than 6.98 MB of memory.

Table X shows the training and inference time consumption of different methods. To comprehensively compare the inference speed of SSRNet and image-based methods, SSRNet takes the whole image as input during the inference stage as

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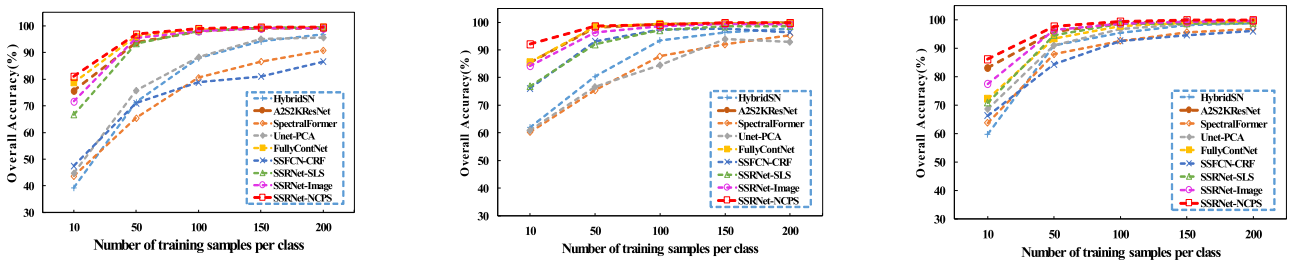


Fig. 12. Overall accuracy of compared methods with different training samples on three datasets. (a) IP dataset. (b) UP dataset. (c) UH dataset.

TABLE X
TRAINING AND INFERRING SPEED OF DIFFERENT METHODS ON THREE DATASETS

Dataset		Patch-based			Image-based			Proposed		
		HybridSN	A2S2KResNet	SpectralFormer	Unet	FullyContNet	SSFCN	SSRNet(S=1)	SSRNet(S=2)	SSRNet(S=3)
IP	Train(s)	166.88	795.76	804.31	334.34	323.45	45.65	326.86	358.09	384.05
	Inference(s)	5.36	6.60	9.56	0.05	0.03	0.16	0.02	0.03	0.03
UP	Train(s)	74.21	556.29	869.40	373.27	775.99	287.08	395.46	450.58	471.47
	Inference(s)	112.79	143.09	52.68	0.09	0.34	1.16	0.19	0.23	0.25
UH	Train(s)	89.39	1329.82	2371.50	476.86	2911.83	1126.10	1026.38	1200.36	1383.81
	Inference(s)	1111.22	1220.64	200.52	0.30	1.07	4.68	0.82	0.91	1.04

same as FCNs. However, it should be noted that in practical applications, the input image can also be divided into multiple small patches for separate calculation to reduce the memory overhead. As shown in Table X, FCNs have a significant speed advantage over the traditional patch-based networks. The inference speed of SSRNet surpasses FullyContNet and SSFCN but is slightly inferior to Unet. The reason why Unet achieving fast inference can be attributed to the use of dimensionality reduction strategy to significantly reduce the computational complexity, while SSRNet achieves the inference speed comparable to Unet without using any dimensionality reduction technique. In addition, since NCPS generates many augmented samples, SSRNet requires a relatively longer training time. Fortunately, the successive rectified learning structure can observably accelerate the model convergence by hierarchical gradient propagation, so the number of training iterations can be appropriately reduced, allowing the overall training time of the network to be kept within an acceptable range.

VI. CONCLUSION

To meet the requirements of hyperspectral classification applications on mobile or embedded platforms, a lightweight FCN with high performance, i.e., SSRNet, is proposed in this study. It achieves fast and high-precision HSIC with extremely low parameters and MAC. In pursuit of a lightweight network, SSRNet is designed to be composed of multiple shallow subnetworks, including an LCA-SFCN and several SASPMs. SSRNet adopts a spatial rectified learning structure and proposes the SASPM by combining depth-wise separable convolution with atrous convolution, which effectively improves the classification accuracy of SSRNet. In order to improve the inference speed while maintaining low memory cost, SSRNet is designed as a patch-wise pixel-to-pixel learning network, and the trained SSRNet can be adapted to receive image input of arbitrary size and fast inference of the full image directly. To address the problem of sparse labels during the training process, an NCPS sampling strategy is also proposed in this study, which enables all spatial locations of the patch to be equally traversed by all labeled pixels through position shift sampling, thus avoiding the spatial information loss in the case of sparse samples case and effectively achieving data expansion of the training samples. The experimental results show that the accuracy improvement brought by NCPS far exceeds that of sparse labeling and full-map training methods in the case of small samples. In addition, an LCA mechanism is also proposed, to effectively improve the accuracy of spectral classification network. However, SSRNet also sacrifices part of the network classification accuracy to pursue the lightweight network structure. In the future, more complex spectral/spatial fully convolutional modules and spatial rectified modules can be designed based on the successive rectified learning structure to achieve higher classification accuracy.

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